

Fundamentals of AI and ML

Lecture 1: Introduction

by

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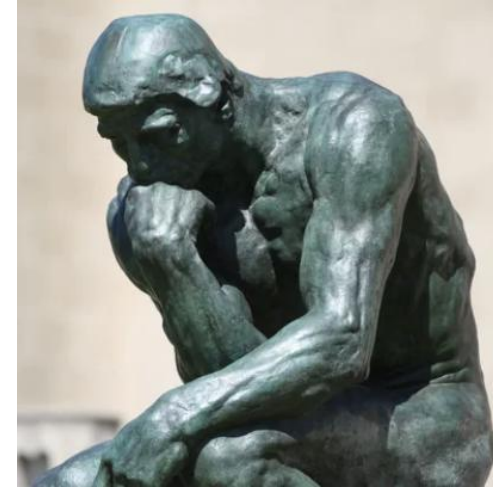
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What is Artificial Intelligence?

Intelligence

- We call ourselves *Homo sapiens* meaning “man the wise” because we are “intelligent”
- For thousands of years, we have tried to understand how we think



But what features/abilities do humans have that are indicative or characteristic of intelligence?

Abstract concepts, mathematics, language, problem solving, memory, logical reasoning, emotions, morality, ability to learn/adapt, etc...

The field of artificial intelligence, or AI, goes further still: it attempts not just to understand but also to **build intelligent entities**

What is Artificial Intelligence?

How do make systems mimic human intelligence?

Four schools of thought

Systems that think
like humans

Systems that think
rationally

Systems that act
like humans

Systems that act
rationally

What is Artificial Intelligence?

Systems that think like humans

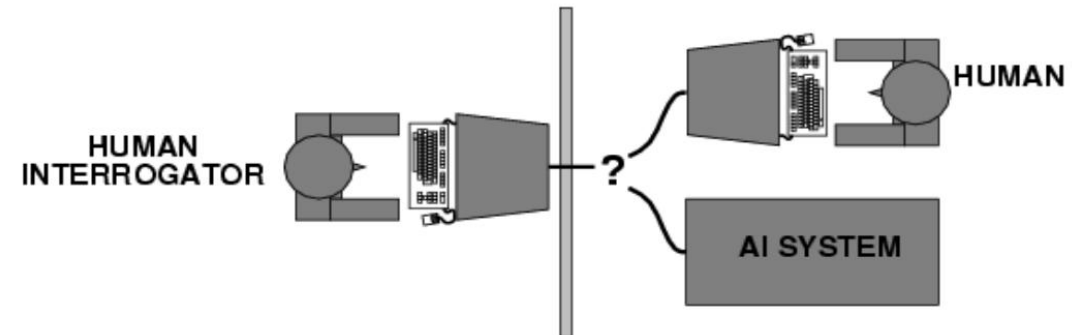
- Cognitive science (**not a part of AI yet**)

Systems that think rationally

- **Aristotle**: What are the correct thought processes?
- Systems that reason in a logical manner
- Systems doing inference correctly

Systems that behave like humans

- Turing (1950) predicted that by 2000 a computer would have a 30% chance of fooling a lay person for 5 minutes
- Suggested major components of AI: **knowledge, reasoning, language understanding, learning**
 - **Knowledge representation**: to store what it knows or hears
 - **Automated reasoning**: to answer questions and draw new conclusions
 - **Natural Language Processing (NLP)**: to communicate
 - **Machine learning**: to detect patterns and extrapolate patterns
 - **Computer vision**: to perceive objects
 - **Robotics**: to manipulate objects and move about



What is Artificial Intelligence?

Systems that behave rationally

- Rational behavior: “doing the right thing”
- Follow rational agent approach
- **Agent:** entity that perceives and acts
- **Rational agent:** acts so to achieve *best outcome*

A focus on rational AI requires learning

- General principles of rational agents
- Components for constructing rational agents



DeepBlue computer chess acting as an agent

Different Sub-Components of AI

Artificial Intelligence

Techniques that equip computers to emulate human behavior, enabling them to learn, make decisions, recognize patterns, and solve complex problems.

Machine Learning

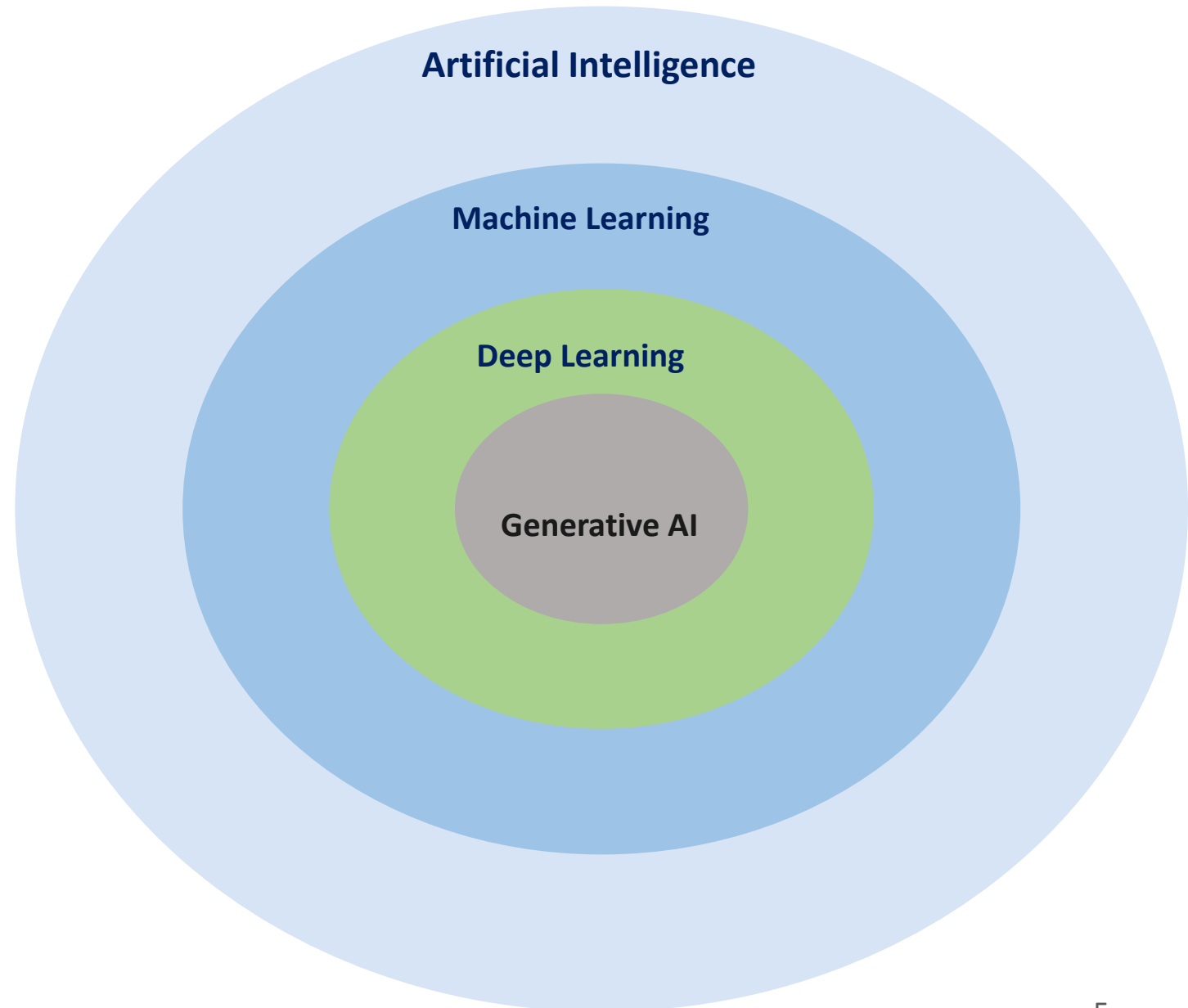
A subset of AI. Uses algorithms to detect patterns in large datasets, allowing machines to learn and adapt.

Deep Learning

A subset of ML which uses neural networks with multiple layers to extract high-level features from raw data.

Generative AI

A subset of deep learning that generates content like text, images, or code based on patterns learned from massive datasets.



What is Machine Learning?

General Definition

“the field of study that gives computers the ability to learn without being explicitly programmed.”



Arthur Samuel

Engineering-oriented Definition

“A computer program is said to learn from *experience E* with respect to some class of *tasks T* and *performance measure P*, if its performance at tasks in *T*, as measured by *P*, improves with experience *E*.”

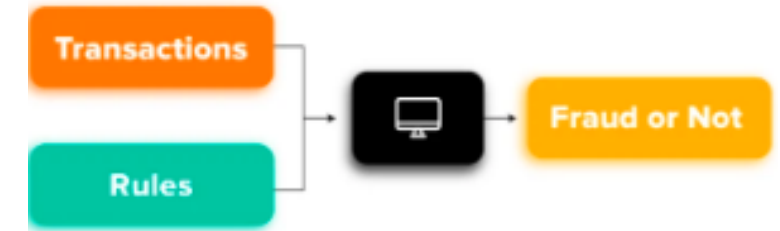


Tom Mitchell

Example — Fraud Detector: T = flagging fraud transactions, E = fraud transactions flagged by customers (training data), P = accuracy of correctly flagged fraud transactions. The detector **improves** as it sees more flagged fraud transactions

An Example of Machine Learning

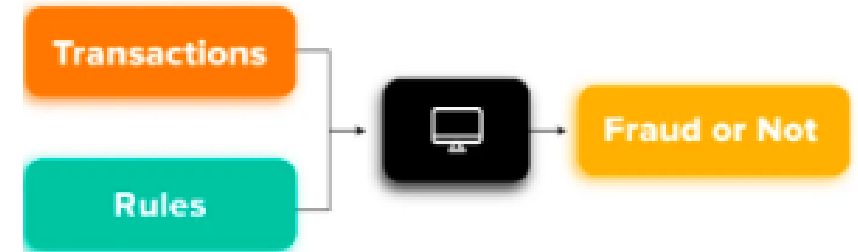
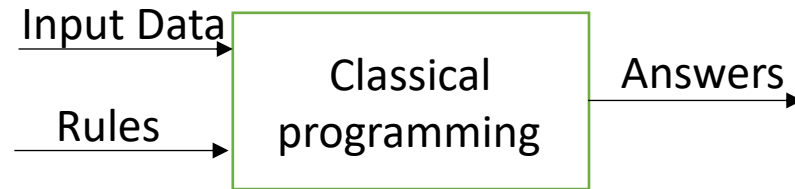
- A *fraud detection system* in banking transactions is an example of ML program where we are given examples of fraud transactions (flagged by customers) and regular transactions



- The examples that the ML program uses to “learn” or “train” is called the **training set (or training data)**
- Each training example is called a **training instance** (or training **sample**)
- The part of the ML system that learns and makes predictions is called the **model**
e.g., a neural network is an example of an ML model
- The **task T** is to flag fraudulent transactions for new transactions
- The **experience E** is the training data
- The **performance measure P** needs to be defined (ex. the ratio of correctly flagged fraud transactions), which is a measure of *accuracy*

Classical Programming vs Machine Learning

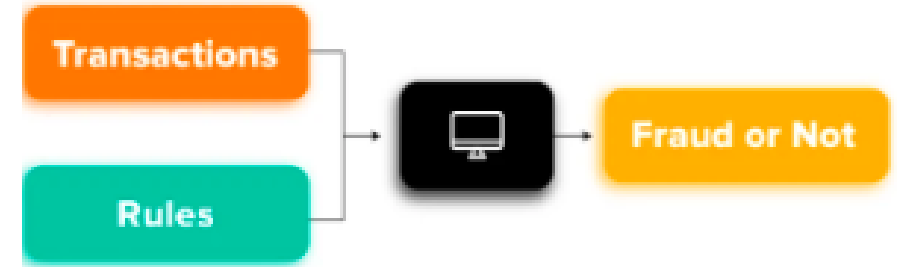
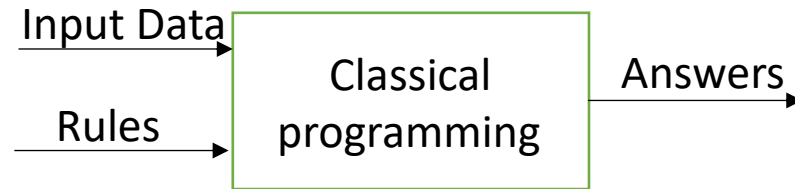
- **Classical programming:** Program/Hard-code the rules for every task



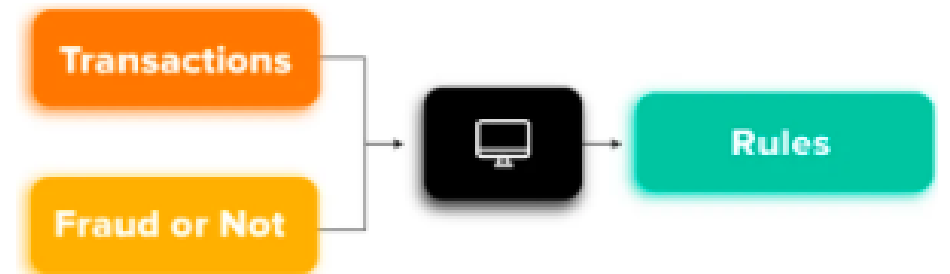
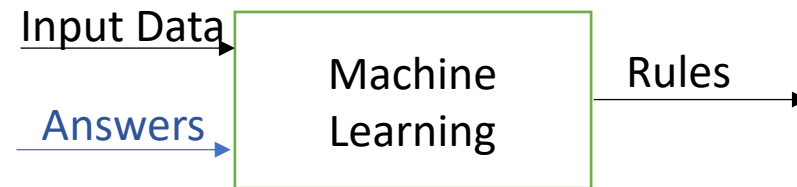
- A human designs the rules to map the input data to answers
- However, for many tasks, it is difficult to manually design and hard-code correct rules
 - detecting spam, fraud transactions
 - recognizing people, objects
 - understanding human speech
 - detecting anomalous behaviors in systems

Classical Programming vs Machine Learning

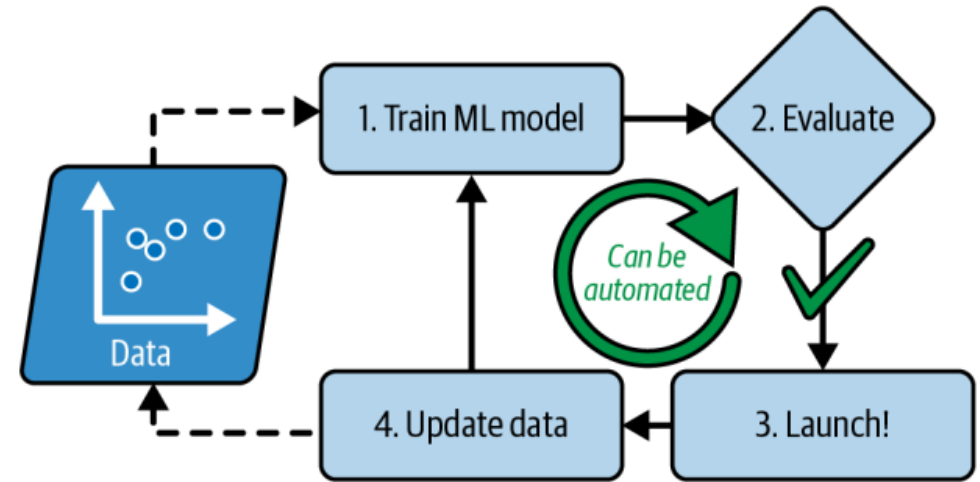
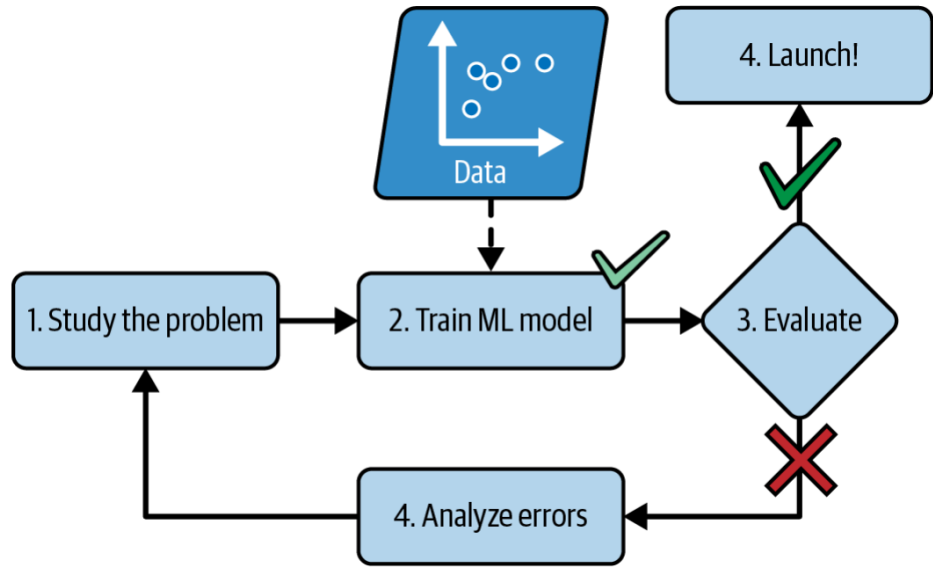
- **Classical programming:** Program/Hard-code the rules for every task



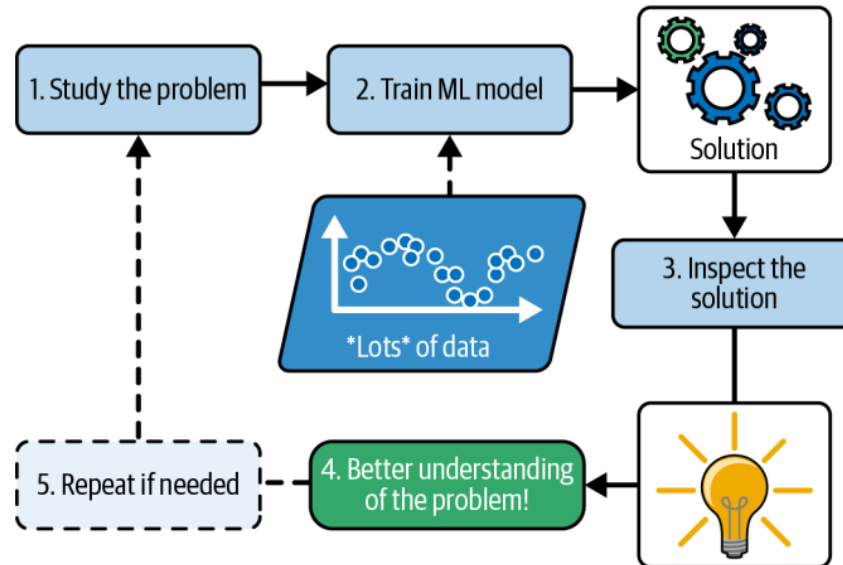
- **Machine learning approach:** An algorithm learn rules automatically from data or experience



Machine Learning Approach



Can automatically adapt to changes



Machine learning can help humans learn (data mining)

Examples of machine learning: Example 1

Automatically diagnosing heart abnormality

- Heart problems influence the electrical activity of the heart. These electrical signals can be measured using electrodes attached to the body (reported in ECG)
- ECG signal gives valuable information about the condition of the heart, which can be used to diagnose the patient and plan the treatment

No abnormalities



Atrial fibrillation



Right bundle branch block



Atrial fibrillation makes the heart beat without rhythm, making it hard for the heart to pump blood in a normal way

Right bundle branch block corresponds to a delay or blockage in the electrical pathways of the heart

Examples of machine learning: Example 1

Automatically diagnosing heart abnormality

- Can we construct a computer program that reads in the ECG signals, analyses the data, and returns a *prediction* regarding the normality or abnormality of the heart?
- **Challenge:** How to design a computer program that turns the raw ECG signal into a prediction about the heart condition?
 - An experienced cardiologist trying to explain his experience to a software developer (which patterns in the data to look for) would not be easy!
- **Machine learning approach:** Teach the computer program through *labelled examples*.
 - Ask the cardiologist (or a group of cardiologists) to *label* a large number of recorded ECG signals with labels corresponding to the underlying heart condition.
 - A machine learning algorithm can then learn to come up with its own rules based on these examples, so that the predictions agree with the cardiologists' labels on the “training” examples
 - The hope is that, if it succeeds on the training data (where we already know the answer), then it should be possible to use the predictions made the by program on previously unseen data (where we *do not* know the answer)

Examples of machine learning: Example 1

Automatically diagnosing heart abnormality

- Ribeiro et. al. (2020) developed a machine learning model for ECG prediction
- 23,00,000 ECG records were used to evaluate the electrical activity of the heart of 17,00,000 patients
- The dataset comes with associated labels for **six different** classes according to the status of the heart, i.e., no abnormalities, atrial fibrillation, RBBB, etc.
- Based on this data, a machine learning model (specifically, a deep neural network model) was trained to automatically classify a new ECG recording without requiring a human doctor to be involved
- To evaluate how the trained model performs in practice, cardiologists with experience in ECG examined and classified
- The average performance of classification was then compared
- The result was that the algorithm achieved better or the same result when compared to the human performance on classifying six types of abnormalities

Examples of machine learning: Example 1

Various concepts central to machine learning can be understood from this ECG example

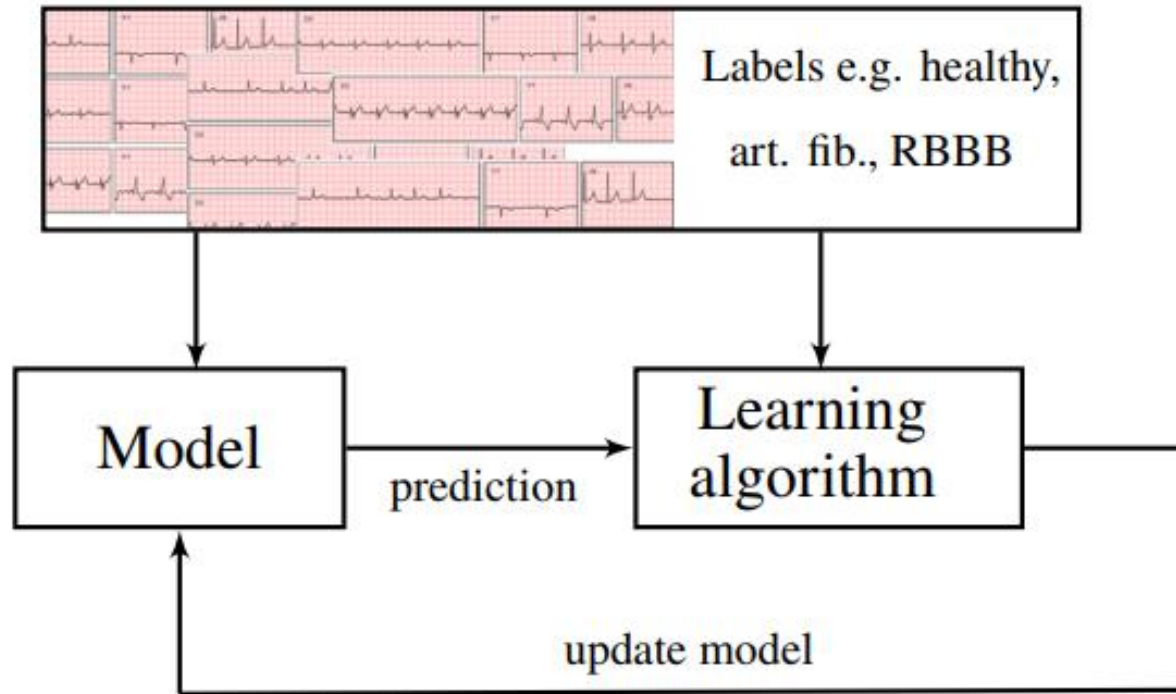
1. Data is an important ingredient in machine learning
2. In the ECG example, each data point consists of an **input** (e.g. ECG signal) and an associated **output** (e.g. label corresponding to the heart condition). Such type of data with both input and output is called *labelled* data
3. “**Training**” a model with labelled data (both inputs and outputs) points is referred to as **supervised learning**
 - Think of the learning as being supervised by the domain expert, and the learning objective is to obtain a computer program that can mimic the labelling done by the expert
4. The ECG example represents a **classification** problem
 - Classification is a supervised machine learning task which amounts to predicting a certain class, for each data point
 - Another type of supervised learning problem is **regression**, where the output is a numerical value
5. In the first phase, a chunk of data is used to train the machine learning model → **Training** data
6. The ultimate goal of the trained ML model is to obtain accurate predictions in future. We say that the predictions made by the model must *generalise beyond the training data*
7. In 2nd phase, new unseen unlabeled data (only inputs) are fed to the computer program to predict the labels

Examples of machine learning: Example 1

Automatically diagnosing heart abnormality

Training phase

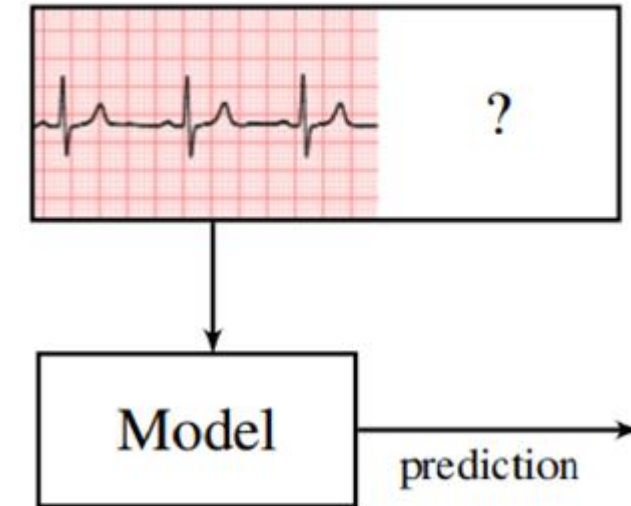
Training data



The parameters of the model are tuned by the learning algorithm such that the model best describes the available training data

Testing phase

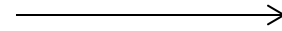
Unseen data



The learned model is used on new, previously unseen data (**test** data), where we hope to obtain a correct classification

Examples of supervised machine learning

- **Supervised learning - classification:** ECG ML model predicted a certain class



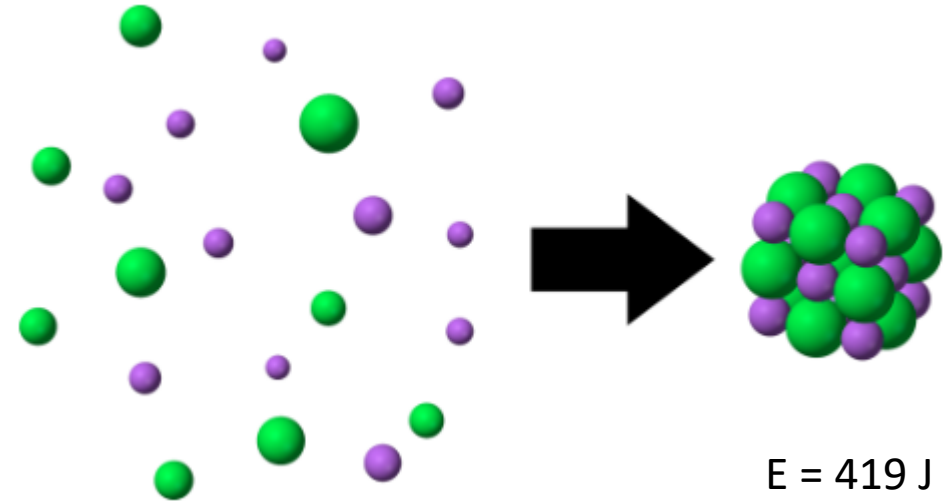
Normal or **Abnormal**?

Examples of machine learning : Example 2

Predicting formation energy of crystals

- **Motivation:**

A basic property of interest when trying to discover or synthesize a hypothetical material is the *formation energy* of a crystal. The formation energy can be thought of as the energy that nature needs to spend to form the crystal from the individual elements. A crystal with lower formation energy is more stably synthesized



- **Challenge:** DFT is very accurate but computationally very expensive, even on modern supercomputers. Hence, only a small fraction of all potentially interesting materials can be analysed
- **Machine learning approach:** Train an ML model that mimics the DFT but is computationally fast
 - Input → Description of the positions and atoms in the candidate crystal
 - Output → Formation energy of the candidate crystal computed using DFT
 - Faber et. al. (2016) used kernel ridge regression to predict the formation energy of 2 million crystals

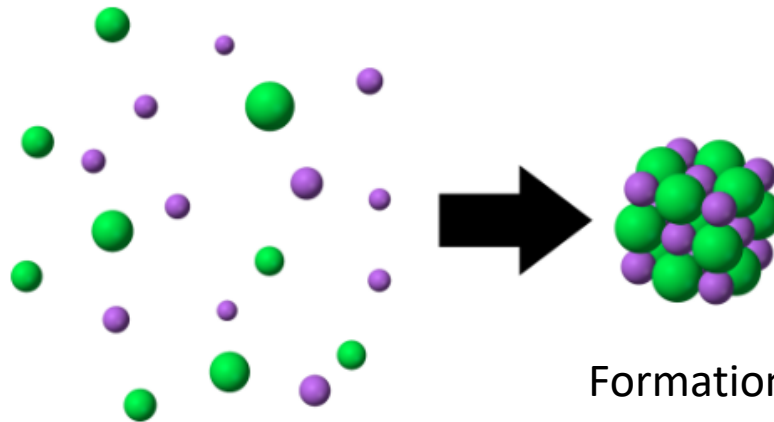
Examples of supervised machine learning

- **Supervised learning - classification:** ECG ML model predicted a certain class



—————→ **Normal** or **Abnormal**?

- **Supervised learning - regression:** Material discovery model predicted a numerical value (formation energy of crystal)



Regression and classification are the two types of prediction problems that we will learn in this training

Three Main Categories of Machine Learning

Supervised

Teacher provides answer



- Labelled data
- Direct feedback
- Predict outcome

- Classification
- Regression

Unsupervised

No teacher, find patterns!



- No labels
- No feedback
- Find hidden structure

- Finding patterns
- Dimensionality reduction
- Anomaly detection

Reinforcement

Teacher provides rewards



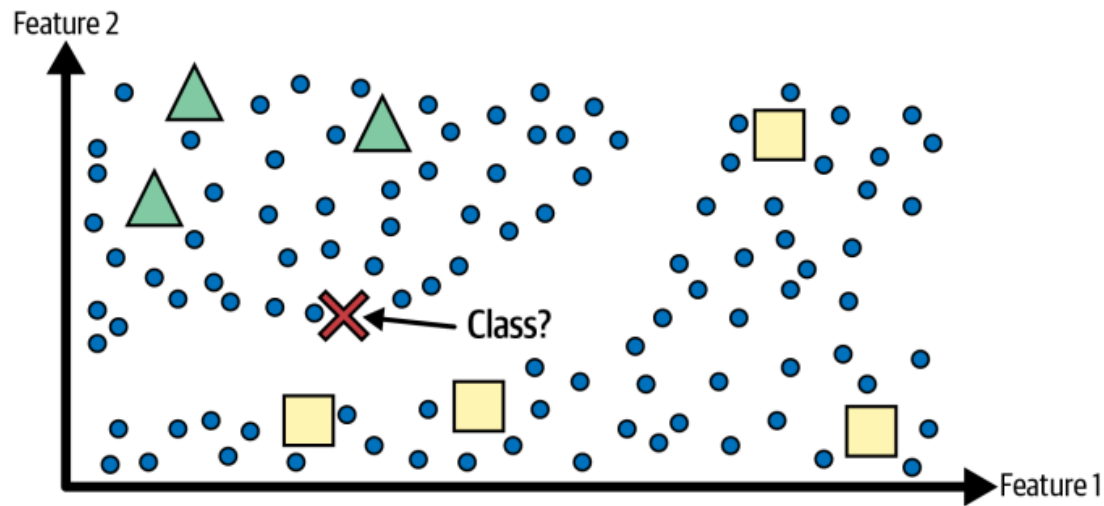
- Decision process
- Rewards
- Learn series of actions

- Gaming
- Control

Two other sub-categories of ML

Semi-Supervised ML

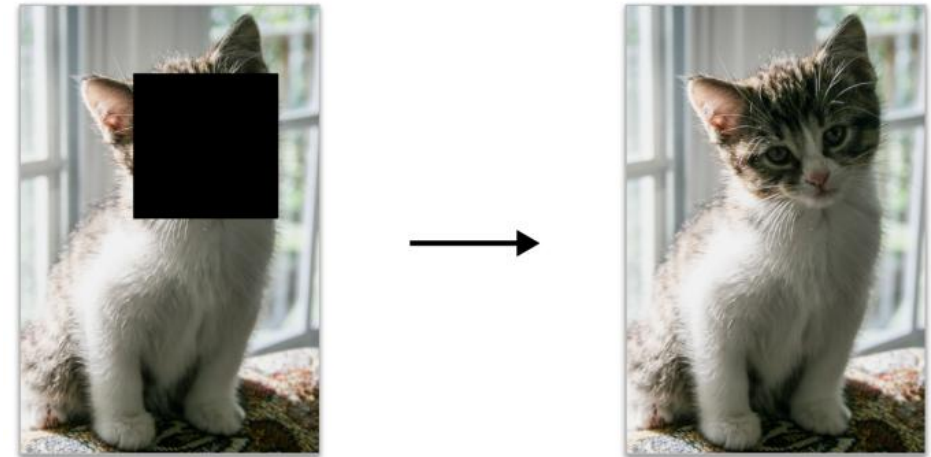
- Labelling data is often **time-consuming**
- Real-world datasets usually contain:
 - Many unlabelled instances (ex. Google photos)
 - Few labelled instances
- These ML algorithms can learn from **partially labelled data**



Semi-supervised learning with two classes (triangles and squares): the unlabelled examples (circles) help classify a new instance (the cross) into the triangle class rather than the square class, even though it is closer to the labelled squares

Self-Supervised ML

- Generates labels from unlabelled data
- Converts a **fully unlabelled dataset** into a labelled one automatically
- After labelling, standard **supervised learning** algorithms can be applied



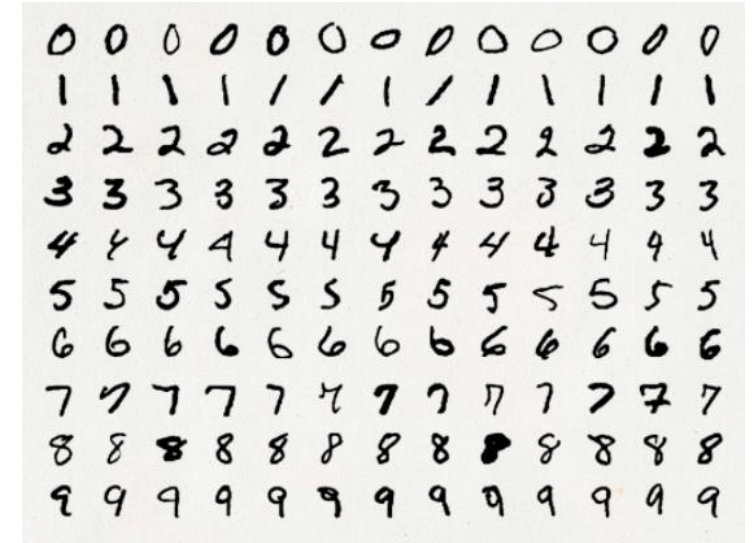
Masked parts of an image are used to train the model to reconstruct the missing regions

Example of Supervised learning

Supervised learning: have labelled examples of what is “correct”

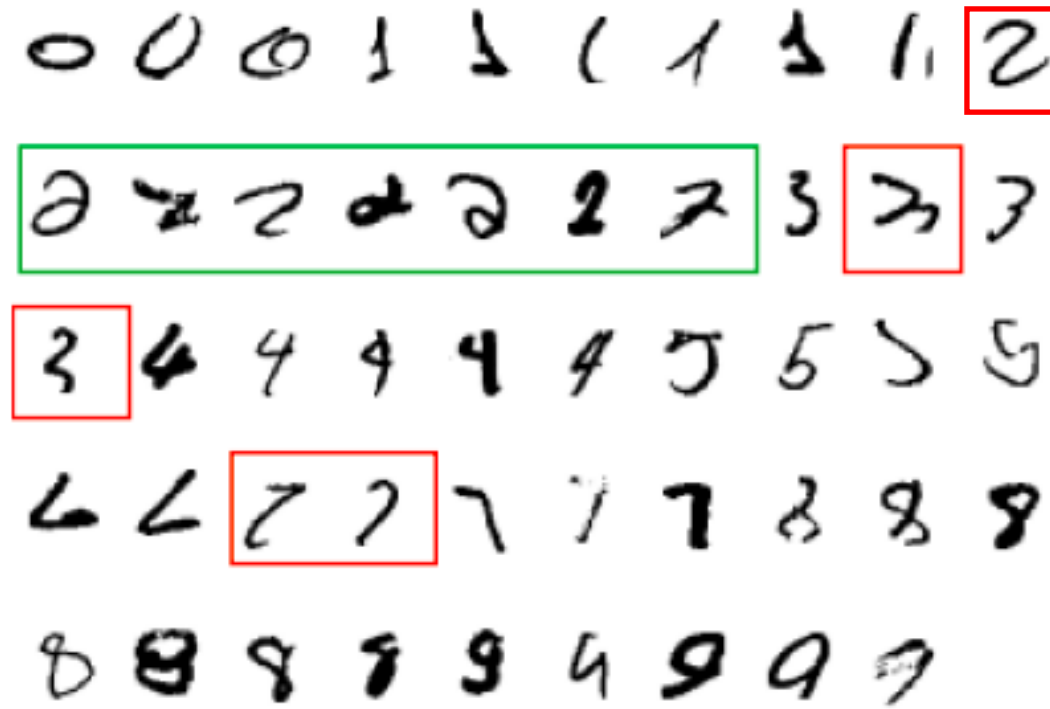
e.g. Handwritten digit classification with the MNIST dataset

- **Task:** Given an image of a handwritten digit, predict the digit class
 - **Input:** a handwritten image of a digit
 - **Output (or target):** the digit class
- **Data:** 70,000 images of handwritten digits labelled by humans
 - **Training set:** first 60,000 images used to train the network
 - **Test set:** last 10,000 images, not used during training, used to assess performance



Example of Supervised learning

Test prediction: What type of images look like a “2”?



Misclassification

Example of Supervised learning

Object Recognition: Detect the class of the object

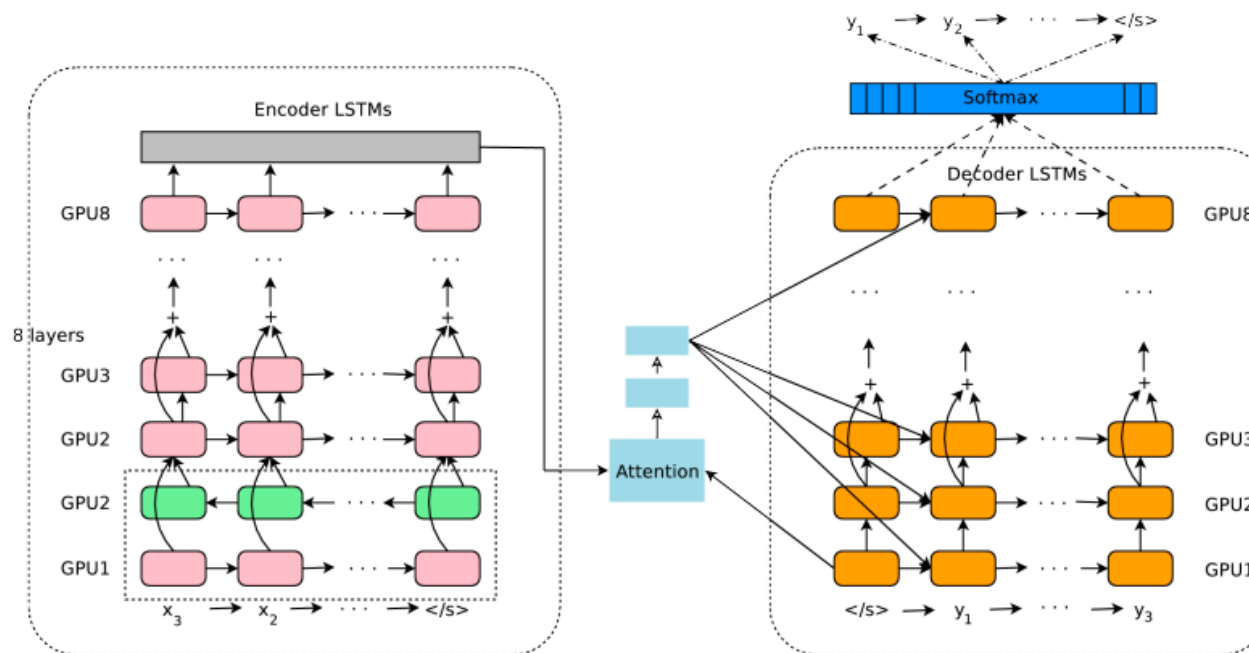
- ImageNet
- 1.2 million labelled images
- 1000 classes
- Lots of variability in lighting, viewpoint, etc.
- Deep neural networks reduced error rates from 26% to under 4%



Example of Supervised learning

Machine Translation: Translate a sentence from one language to another (**Google Translate**)

Google
Neural
Machine
Translation
(GNMT)



PBMT – Phrase-based
Machine Translation

GNMT has been
replaced in 2020 by
Transformer + LSTM
model

<i>Input sentence:</i>	<i>Translation (PBMT):</i>	<i>Translation (GNMT):</i>	<i>Translation (human):</i>
李克強此行將啟動中加總理年度對話機制，與加拿大總理杜魯多舉行兩國總理首次年度對話。	Li Keqiang premier added this line to start the annual dialogue mechanism with the Canadian Prime Minister Trudeau two prime ministers held its first annual session.	Li Keqiang will start the annual dialogue mechanism with Prime Minister Trudeau of Canada and hold the first annual dialogue between the two premiers.	Li Keqiang will initiate the annual dialogue mechanism between premiers of China and Canada during this visit, and hold the first annual dialogue with Premier Trudeau of Canada.

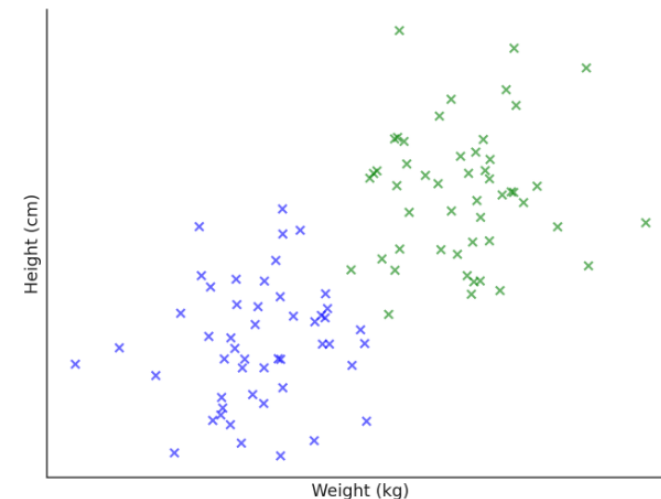
Example of Unsupervised learning

Unsupervised learning: no labelled examples, you only have input data. You are looking for interesting patterns in the data

- To find clusters in data
- To find a compressed representation
- To find a generative model that could be used to generate more data

E.g. Clustering – Group the input data into separate classes

	Input 1 ↓	Input 2 ↓
	Height (cm)	Weight (kg)
example 1	168	60
example 2	165	56
	⋮	⋮
example N	181	97

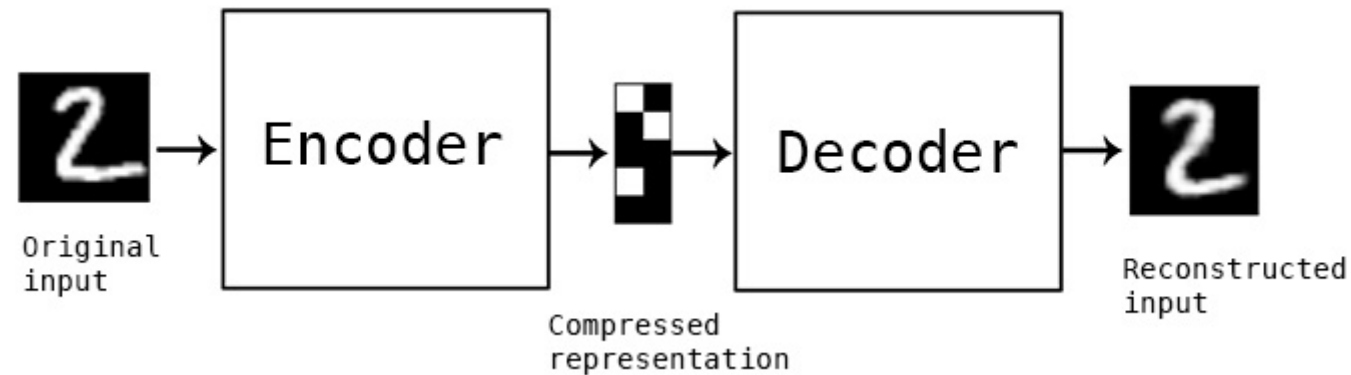


Example of Unsupervised learning

Unsupervised learning: no labelled examples, you only have input data. You are looking for interesting patterns in the data

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- To find a generative model that could be used to generate more data

E.g. Compressed representation – Find a reduced dimension of the input



Example of Unsupervised learning

Unsupervised learning: no labelled examples, you only have input data. You are looking for interesting patterns in the data

- In **generative modeling**, we want to learn a distribution over some dataset, such as natural images. We can then sample from the generative model and see if it looks like the data.

Generated faces
(not true faces)



Example of Unsupervised learning

Unsupervised learning: no labelled examples, you only have input data. You are looking for interesting patterns in the data

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GANs, VAEs, Diffusion

Example of Unsupervised learning

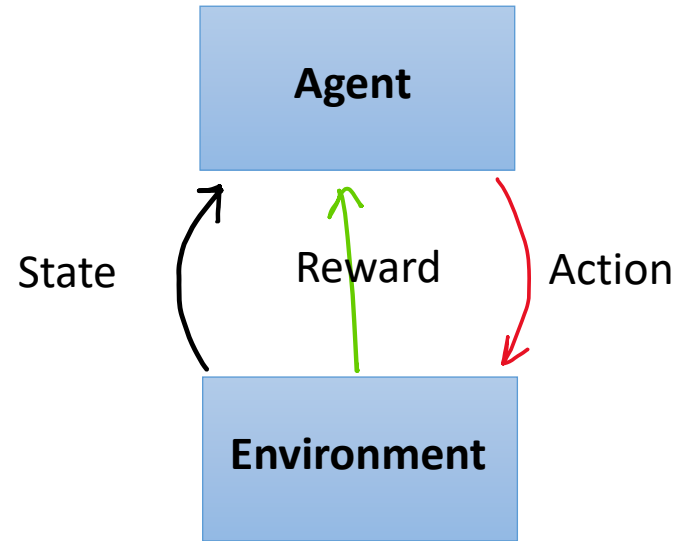
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GANs, VAEs, Diffusion

Example of reinforcement learning



Goal: Learn to choose actions that maximize rewards

- An **agent** (e.g. player) interacts with an **environment** (e.g. enemy-killing game)
- In each time step
 - the agent receives observations of the **state** (e.g. how many enemies remaining)
 - the agent picks an **action** (e.g. moving to safe location, or killing an enemy)
- The agent will periodically receive some **rewards** (e.g. health, ammunition, scores)

Idea of RL is based on animal psychology

- Reinforcements are used to train animals

- Negative reinforcements

- Hunger
- Pain

- Positive reinforcements

- Food
- Pleasure



- This psychology is applied here to computers

- Rewards: numbers or numerical signals indicating how good the agent performed
- Example of rewards: Win/loss in games, points earned, etc.

There are many RL applications

- Self-driving cars to follow lanes



- Board games: Alpha Zero (in chess)
- Digital games
- Education (using ChatGPT, Alexa, etc.)



Supervised Learning: Background

- We start with **Supervised Learning**
- The task is to **learn the function** f that best maps certain **input** ($\mathbf{x}$) to **output** (y)

		A	B	C
	1	x1	x2	y
Sample 1	2	2.1	1.5	3
Sample 2	3	2.3	-0.6	1
Sample 3	4	3.1	0.9	3.2
Sample 4	5	2.4	-0.1	1.2

		A	B	C
	1	x1	x2	y
	2	2.1	1.5	Cat
	3	2.3	-0.6	Dog
	4	3.1	0.9	Cat
	5	2.4	-0.1	Dog

$$y = f(\mathbf{x})$$

- We don't know what the function (f) looks like or its form
 - If we knew the form, we would use it directly and we would not need to learn it from data
- Moreover, the output y is often observed with some errors e that is independent of the input $\mathbf{x}$
 - The error could be due to measurement instrument errors
 - The error could be due to not including enough input features to sufficient characterize the mapping from $\mathbf{x}$ to y

$$y = f(\mathbf{x}) + e$$

Supervised Learning: What do we do in supervised ML?

- The task in supervised ML is to **learn the function** f that best maps certain **input** ($\mathbf{x}$) to **output** (y)

$$y = f(\mathbf{x})$$

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$$y = f(\mathbf{x}) + e$$

- In supervised ML, we use **labelled training data** (input-output pairs) that contains examples of how some *input* $\mathbf{x}$ relates to *output* y to learn the input-output mapping
 - Say, N examples of labelled training data: $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)$
 - Labelled data: Each input $\mathbf{x}_i$ is accompanied by an associated label y_i , which be jointly recorded or labelled later by some domain expert

Supervised Learning: What is reason for learning a function?

$$y = f(\mathbf{x}) + e$$

- The most common type of ML is to learn the mapping $y = f(\mathbf{x})$ to **make good predictions** of the output for new examples of input (say, $\mathbf{x}$) → to **generalise** well beyond training data
 - This is called **predictive modeling** or **predictive analytics** and our goal is to make the most accurate predictions possible
 - Hence, we are not really interested in the form of the function (f) that we are learning, only that it makes accurate predictions
- We could learn the mapping of $y = f(\mathbf{x})$ to **understand more about the relationship** in the data → **statistical inference**
 - If this were the goal, we would use simpler methods and give more importance to understanding the learned form of f than making accurate predictions
 - E.g. “Does eating seafood increase life expectancy?” requires careful reasoning about the function that was learned

Supervised learning: Two types of data

The variables contained in the data (input $\mathbf{x}$ as well as output y) can be of two different types:

- **Numerical (quantitative)**
 - Has a **natural ordering**, i.e., a numerical variable maybe larger or smaller than another one
 - Can be **continuous** or **discrete**
- **Categorical (qualitative)**
 - Lacks a natural ordering
 - Is **always discrete**
- The notion of categorical vs. numerical applies to both the output variable y and to the p elements x_j of the input vector variable $\mathbf{x} = [x_1 \ x_2 \ \cdots \ x_p]^T$
- Also, the p components of the input vector do not have to be of the same type and can be a mix of categorical and numerical input
- However, the output y is either categorical or numerical

Numerical vs Categorical data: Examples

Data Type	Example	Handled as
Number (continuous)	15.58 km/h, 11.50 km/h	
Number (discrete) with natural ordering	0 bikes, 1 bikes, 2 bikes	
Number (discrete) <i>without</i> natural ordering	1 = Argentina, 2 = Brazil, 3 = India	
Text String	Hello, Bye, Welcome	
?	$3.4 + 5.6i$, $-6.2 + 0.1i$	

- The distinction between numerical and categorical is sometimes arbitrary
- For example, having no bike is qualitatively different from having bikes, and we can use the categorical variable 'bikes: yes/no' instead of the numerical '0, 1 or 2 bikes'
- Therefore, the **decision lies the ML engineer** whether a certain variable is to be considered as numerical or categorical

Supervised ML: Regression vs Classification

- Output variable y ? → **categorical** → **Classification**
- Output variable y ? → **numerical** → **Regression**
- Note that the p – dimensional input vector variable $\mathbf{x} = [x_1 \ x_2 \ \cdots \ x_p]^T$ can be either numerical or categorical for both regression and classification problems
- It is **only the type of the output** that determines whether a problem is a regression or a classification problem
- Classification: **Binary** vs **Multi-class**
 - Output is categorical
 - Output can take values in a finite set
 - **Binary** classification, if only two set of values. e.g. True or False, 0 or 1, -1 or 1
 - **Multi-class** classification: if more than two set of values. e.g. “Sweden”, “Norway”, “Finland”, “Denmark”

Examples of classification and regression

Problem	Input	Output	Classification or Regression?
Spam detection			
Stock price prediction			
Speech recognition			

Examples of classification and regression

Problem	Input	Output	Classification or Regression?
Digit recognition			
Housing valuation			
Weather prediction	Sensor data (images, wind speed)		

The Machine Learning Workflow

Step 1: Define the Problem

What decision to automate? Is there a learnable pattern? Do I have data to learn from?

Getting this wrong wastes everything that comes after.

Step 2: Collect and Clean Data

Gather examples, fix errors, handle missing values.

In practice, this is often the most time-consuming step; real-world data is messy.

Step 3: Prepare Features

Transform raw data into a form the algorithm can use.

You transform it — maybe normalising numbers, encoding categories — into a form the model can actually work with

Step 4: Train the Model

This is where the algorithm runs: The algorithm finds patterns by fitting to training data.

Step 5: Evaluate Performance

You test the model on data it hasn't seen before. Is it accurate enough? Is it fair? Does it generalise, or did it just memorise the training set?

Step 6: Deploy and Monitor

You put the model to work in the real world, but you don't walk away. Models can degrade over time as data patterns shift, so monitoring is essential.

These six steps are iterative: you'll often loop back to earlier stages as you learn more

How it works: Predicting Schedule Delay

Problem: Can we predict whether a construction activity will overrun its planned date?

Step 1

Collect data: Pull 500 past records from project history: planned duration, crew size, site complexity, weather risk, subcontractor tier. This becomes the training data.

Step 2

Label each row: “Ran late” column: Yes / No. This is our label — what we want the model to learn to predict.

Step 3

Train the model: Feed 400 rows to a learning algorithm. It finds patterns: e.g. high complexity + small crew → often late finish.

Step 4

Test on new data: Show the model the remaining 100 rows it has never seen. Suppose it correctly predicts 82 of them.

Step 5

Use in practice: For each upcoming activity, input its features. The model outputs a delay risk score. Project Manager is alerted if risk > 70%.

Result: An early-warning system — weeks before delay would have been noticed. No new data collection needed; Bechtel already has these records.

What Data Looks Like: Features and Labels

Think of it as a spreadsheet. Each row = one project activity. Each column = one piece of information.

Activity ID	Size (m²)	Duration (days)	Complexity	Crew Size	Ran Late?
A-001	12,000	180	High	25	No
A-002	8,500	210	Medium	18	Yes
A-003	21,000	150	Low	40	No
A-004	5,000	300	High	12	Yes

Features — the inputs the model uses to learn

Label — what we predict

Feature: a measurable property of each example. **Label:** the answer we already know and want the model to reproduce.

Key Concept: Bias and Variance

The goal of any supervised ML algorithm is to achieve both low bias and low variance.

Bias

How wrong the model is on average

Analogy

A spirit level that always reads 2° off — it is systematically wrong regardless of who uses it.

- High bias → model is too simple, misses real patterns (underfitting)
- Low bias → model makes few systematic errors

Variance

How much the model changes with different training data

Analogy

A junior engineer who gives wildly different cost estimates depending on which past projects they happen to remember.

- High variance → model memorises training data, fails on new data (overfitting)
- Low variance → model is stable across different samples

Goal of supervised ML: achieve LOW bias AND LOW variance. In practice, reducing one often increases the other: this is the bias-variance trade-off.

What Can Go Wrong?

Bad data → bad model

If historical project records have errors, missing entries, or biases, the model learns those flaws. Garbage in, garbage out.

Overfitting

The model memorises the 400 training activities but fails on new ones. Like a student who memorises past exam papers but cannot answer new questions.

Distribution shift

World changes: a model trained on 2018–2022 projects may not reflect post-pandemic supply chains or new material costs.

Biased predictions

If certain site types or contractors are underrepresented in training, predictions may systematically disadvantage them.

Summary — What to Remember

1

ML lets computers learn rules from data — you provide examples, not explicit instructions.

2

AI is the broad field; ML is a subset that uses data. Deep Learning and Generative AI are nested subsets.

3

Supervised learning (our main focus) requires labelled data: features and a label.

4

The workflow: define the problem, collect data, prepare features, train, evaluate, deploy.

5

Every ML model can fail. Bad data, overfitting, and distribution shift are the three most common causes.

6

Bechtel has immediate applications: delay prediction, safety monitoring, cost forecasting.